



CTrees: Journey to Cloud-Native GIS

*TNC - Beginner Data Science
Training*

February 2026
Naomi Provost, Head of Engineering

About CTrees

A science nonprofit
**tracking carbon in every
living tree** to accelerate
natural climate solutions.

30+ scientists, data engineers, and policy experts

20+ years of experience building global carbon monitoring systems and solutions



CTrees team at 2024 staff retreat at UCLA (top);
Sassan Saatchi, CTrees' CEO and chief scientist (bottom)

About Me

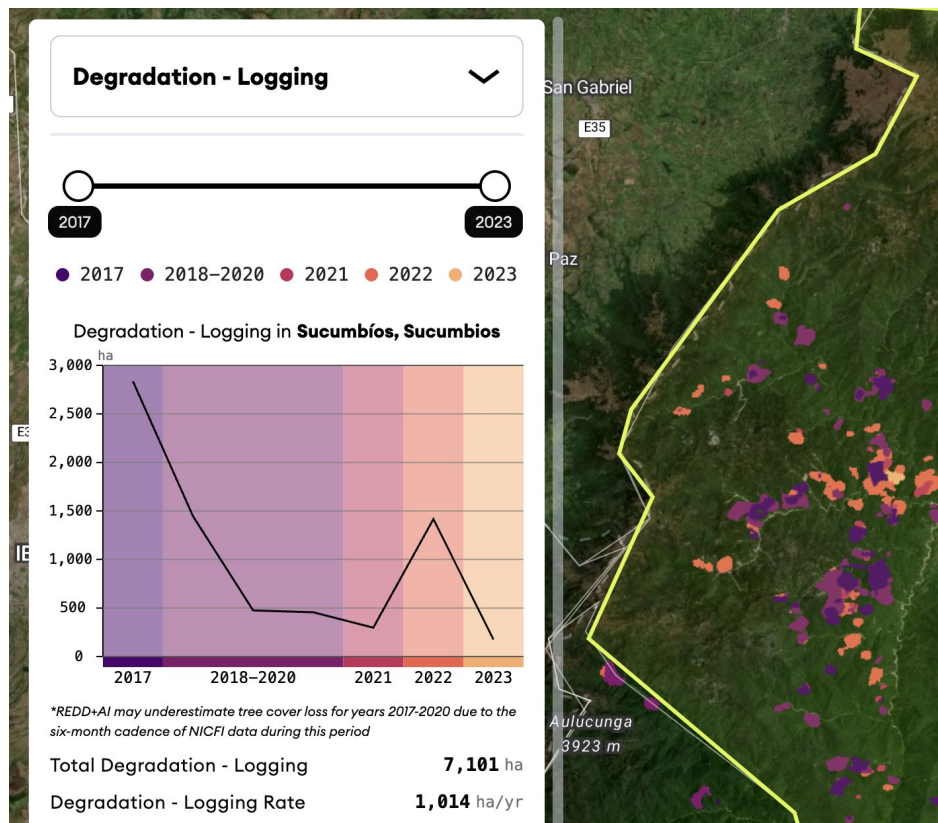
- UW Geography 2016
- GIS Analyst / Data / Software Engineer
- Microsoft | 5-person Startup
- Joined **CTrees** in **2023**
- **Technical Product Lead (2023-2025)**
- Head of Engineering (2025 - Present)



Core Datasets

High-resolution, Global time-series GeoTIFFs:

- Above Ground Biomass
- Tree Cover Loss
- Degradation: Fire, Logging, Roads
- Soil Carbon
- Tree Height
- Forest Productivity



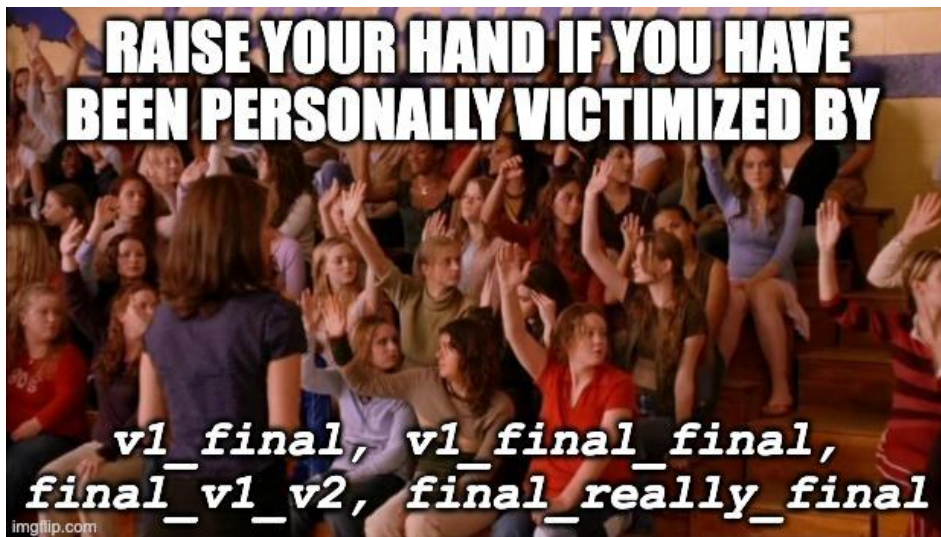
Major Challenges

Data Management

- Manual, error-prone dataset versioning
- Inconsistent file and folder structures
- Missing or incomplete metadata

Analysis Limitations

- GDAL cmd / R / QGIS
- Not cloud-native compatible
- No standardization of inputs



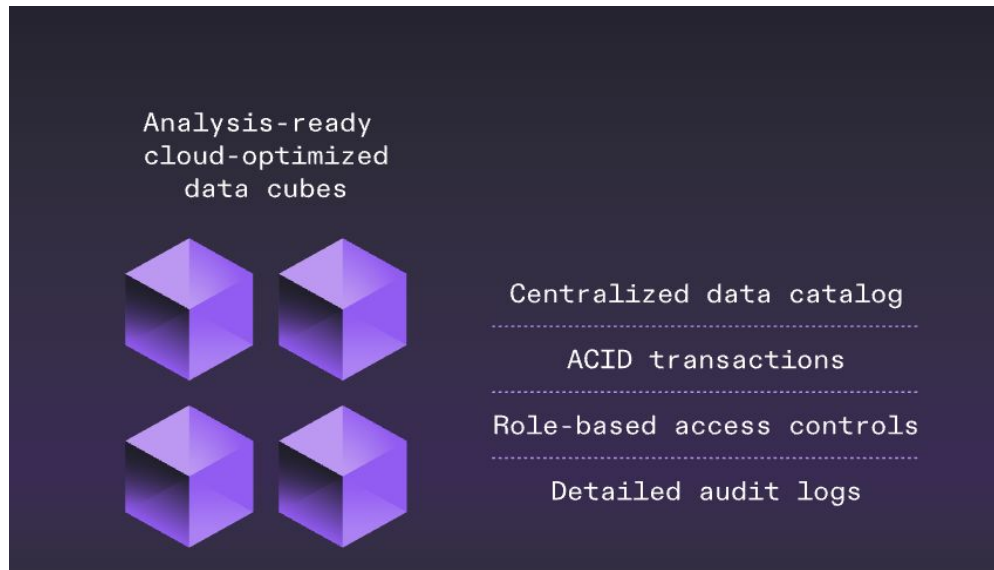
Background Concepts (Basic)

-  **Zarr** = N-dimensional data cubes stored with metadata
-  **Icechunk** = Zarr with git-like version control
-  **xarray** = Python library for raster analysis / manipulation
-  **dask** = Python library used to enable cloud-native data processing

Join the intermediate / advance sessions to learn more about these in practice.

Data Management Solution

- **Icechunk**
 - Versioning embedding
 - Metadata embedding
 - Time series data cubes
 - Cloud native
- **Earthmover's Arraylake**
 - Data Catalog
 - WMS
 - API Access

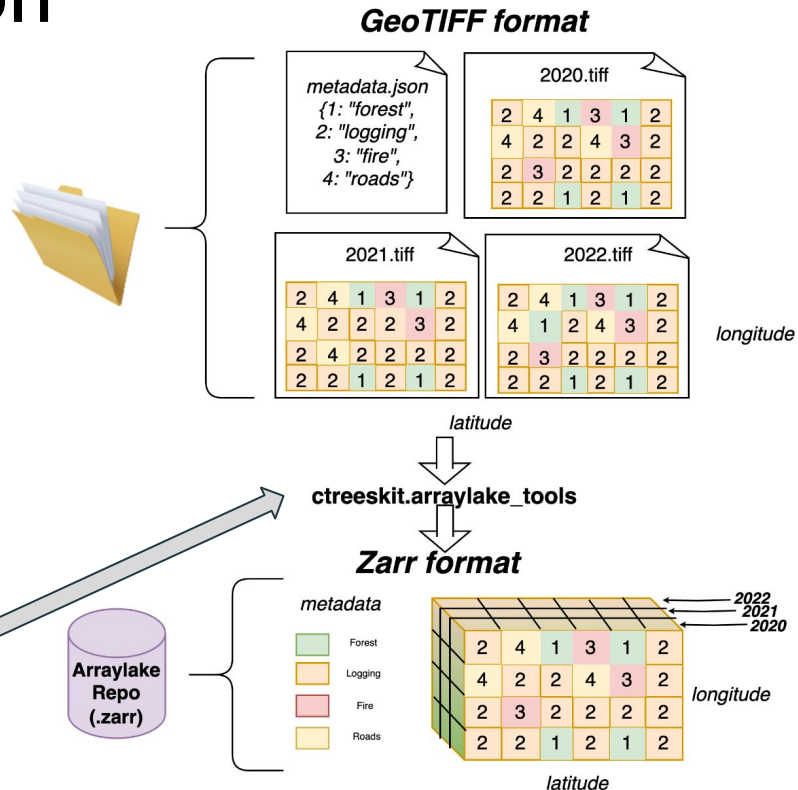


Want to learn more about [Earthmover](#)? Join the intermediate session next week!

Data Management Solution

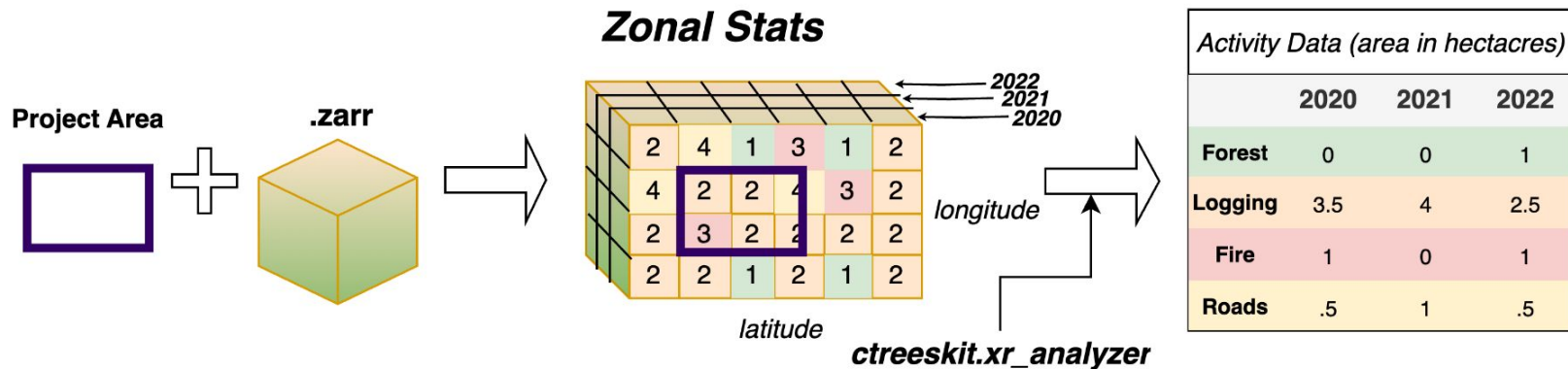
- CTrees' engineering team developed [ctreeskit](#), an open-source pip package to convert geotiffs to Icechunk

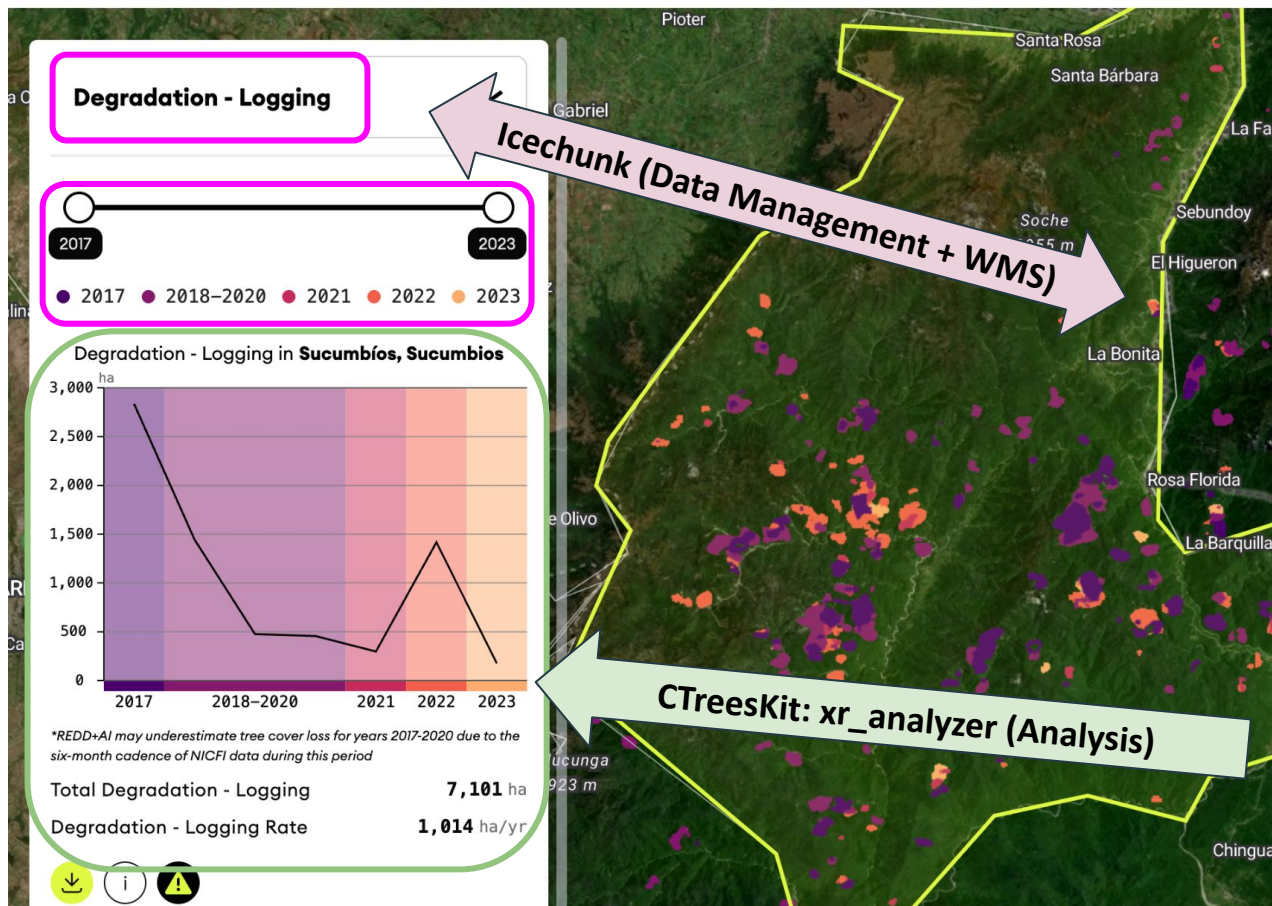
```
repo_config.json
{"repo_name": "example", "s3_path_prefix":
"s3://test/data", "s3_path_suffix":
"annual.tif", "date_range": [2000, 2023],
"classification": {1: forest, 2: non_forest"}}
```



Types of Analysis (now cloud-native using CTreeskit!)

- Process geometry (geometry area)
- Clip by geometry
- Reproject & match datasets
- Precise area conversion
- Merge multiple classification datasets
- Calculate area zonal stats





Alternative Solutions?

STAC (SpatioTemporal Asset Catalog)

- Strong standard for **discovering and organizing geospatial files**
- Best suited for **file-based workflows** (COGs, Zarr on object storage)
- Analysis typically happens **outside the catalog**
- Perfect for when data is in different CRS or level-1 data

Why we didn't choose STAC

- Not designed for **scalable, low-latency analytics** over large multidimensional datasets
- Requires coordinating many files for analysis, increasing **operational complexity**
- Most of our datasets are level-3 analysis ready dataset which can be mosaiced and merged into data cubes

Why Icechunk / Arraylake instead

- **Native multidimensional arrays** with server-side computation
- Supports **fast, on-demand analysis** for product APIs

Other Benefits

Science → Data

- No more throttling issues when pipelines are ran
- Inputs into pipelines are clearly structured, code is simplified
- Standardized functions
- Easier to share data amongst other teams
- Easy version cleanup (thanks for Arraylake)

More information

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